

# Natrosensor: A Python/Django browser-based platform for electrochemical biosensor workflows

Joreyn Angelo T. Tagaza<sup>1\*</sup>, Resmond L. Reaño<sup>1,2\*</sup>, and Donna Wren Libunao-Bueta<sup>1\*</sup>

<sup>1</sup> Biomedical Technology and Devices Laboratory, Department of Engineering Science, College of Engineering and Agro-industrial Technology, University of the Philippines Los Baños, Laguna, Philippines<sup>ROC</sup> <sup>2</sup> Pathology and Diagnostics Division, Zoonoses Center, University of the Philippines Los Baños, Laguna, Philippines ¶ Corresponding author \* These authors contributed equally.

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## Software

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## Summary

Antibiotics in aquatic environments pose severe health risks by promoting antimicrobial resistance. To help address this concern, the ‘Natrosensor’ platform was developed as a Python/Django browser-based software supporting electrochemical biosensors workflows. The software integrates Python modules and scientific computing libraries to perform complex data analysis, including logistics model calibration curve fitting, concentration prediction, and statistical validation. Utilizing an SQLite3 database, the platform enables centralized data storage, longitudinal tracking, and comparative trend analysis across multiple sampling locations. The application features a responsive frontend for multi-device access, which was tested via localhost and can be deployed securely via an Apache server with WSGI.

## Statement of need

While the electrochemical device is the “heart” of the detection system, its full potential is realized through integration with software to perform signal processing and conversion to concentration reading (Letaief et al., 2019). Four-parameter logistics (4PL) regression was applied to calculate the concentration from sensor signals that exhibit non-linear ligand-receptor relationships (Rajkumar et al., 2025). Centralized cloud-based data storage allows tracking of targets across different geographic locations. It also ensures remote sampling and data collection in real-time (Nuthalapati et al., 2024; Subbarao et al., 2024).

However, most available software offers basic visualization but with limited analytical tools for tailored and precise calculations. Moreover, data is often stored locally, disabling regional monitoring and data sharing. Lastly, the traditional laboratory-based detection techniques, often tied to complex IoT systems, are too expensive or difficult to adopt by developing countries (Chhajed et al., 2023; Lim et al., 2010).

## State of the field

The Natrosensor platform took advantage of the Python/Django framework in creating high-level features for enhanced user experience. The stylish interface is user-friendly and is accessible even in remote areas through the internet connection or local host deployment. Extension features, such as user authentication, geolocation, sampling schedules, and record monitoring, strengthen its position as a genuine support to a Point-of-care device.

## Software design

Django 5.0.1 with a SQLite3 database backend enabled combined scientific computation and modern web technologies to deliver real-time biosensor data processing and visualization capabilities. This includes application of various programming languages to deliver exceptional features necessary for its intended application.

The software consists of the 'core' configuration module and the Natrosensor backend calculation module. The 'core/setting.py' is the central configuration file containing instructions for the database, middleware, static file paths, email backend, and security parameters. The 'natrosensor/models.py' consists of the data models used in the database design. The 'natrosensor/views.py' contains the functions for implementing the business logic for all application environments. The 'natrosensor/urls.py' instructs the URL routing configuration mapping the endpoints to view functions. Lastly, the 'natrosensor/forms.py' handles the form classes for user input validation.

The software pages are built using Tailwind CSS, HTMX, and Alpine.js. The HTMX utilizes a JavaScript library that enables AJAX requests, CSS transitions, and WebSocket communication without writing JavaScript. It reduces the page's time to reload, thus improving the transition speed. Alpine.js is used to build the interactive components.

The database uses a user-record-event entity-relationship model to map who initiates the action, the object of concern, and the specific action. A custom 'user model' consists of fields for email address as the unique identifier, the first and last name, and the user's affiliation. The 'records model' serves as the storage of the biosensor data, including sample name, antibiotic type, trial number, environmental parameters, geographic location, antibiotic dose, electrochemical reading, and graph images. The 'event model' manages the user scheduling with fields for event name, date, time, and details. Lastly, an 'OTP model' is included, handling the OTP authorization with hashed codes and expiration timestamps.

The user authentication system generates a 6-digit OTP sent via an official email configured in 'settings.py'. The user should enter the OTP upon receiving it to verify the email address. Upon successful completion, the user account is created with email as the username and a user-nominated password.

## Research impact statement

The integration of computational methods, such as curve fitting and prediction algorithms transforms traditional electrochemical measurements into intelligent, self-calibrating systems. Future direction includes the integration of AI and ML algorithms to achieve a comprehensive calculating platform that can be scaled up from a pilot in a single laboratory with few users to regional monitoring networks accessed by multiple institutions and with distributed sampling sites, without requiring infrastructure redesign. The software and the hardware application can be scaled from antibiotic detection to various molecules because of the flexibility of the electrochemical biosensor technology.

## Mathematics

The software's core analytical framework uses a 4PL model defined in [Equation 1](#).

$$y = d + \frac{a - d}{1 + \left(\frac{x}{c}\right)^b} \quad (1)$$

where  $a$  is the minimum dose response,  $d$  is the maximum dose (sensor saturation) response,  $c$  is the half-maximal effective concentration ( $EC_{50}$ ), and  $b$  is the Hill's slope describing the

steepness of the dose-response curve. The fitted model is evaluated using the  $R^2$  value to quantify the goodness-of-fit using Equation 2.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

The limit of detection ( $LOD$ ) and the lower limit of quantification ( $LLOQ$ ) in terms of the response are computed using Equation 3 and Equation 4. The  $LOD$  and  $LLOQ$  in terms of the antibiotic dose are calculated using the inverse 4PL equation shown in Equation 5.

$$LOD = \bar{x}_{blank} + 3 \cdot SD_{blank} \quad (3)$$

$$LLOQ = \bar{x}_{blank} + 10 \cdot SD_{blank} \quad (4)$$

$$x = c \left( \frac{a - y}{y - d} \right)^{\frac{1}{b}} \quad (5)$$

## AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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